Maths Chapter 6

Population : Entire group we want to study · Ex: all students in a uni

Parameters : μ = population mean

σ^2 = population variance

Sample: A smaller subset of the population.

Parameters : $\bar{X}$ = sample mean

S^2 = sample variance

Formulae :

① $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ and $E[S^2] = \sigma$

② $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(X_i - \mu)^2$

③ $Var(\bar{X}) = \frac{\sigma^2}{n}$

↳ variance of Sample Mean : how much the sample mean changes across multiple samples.

Central Limit Theorem (CLT)

For a set of independent RVs : $X_1, X_2, \ldots, X_n$ with the same mean (μ) and variance (σ^2), then for large n, the sum $X_1 + X_2 + \ldots + X_n$ is normally distributed even if the RV's themselves aren't normal.

Let $Y = X_1 + X_2 + \ldots + X_n$ $\quad Y \sim \text{Gaussian}(n\mu, n\sigma^2)$

$$E[Y] = n\mu$$

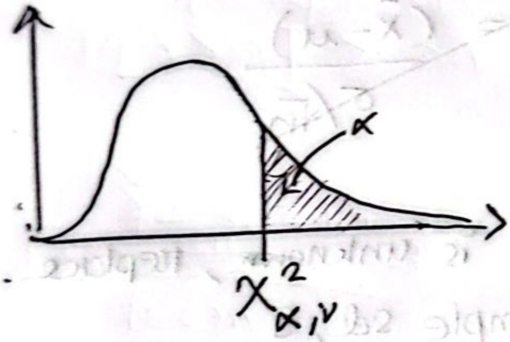
$$Var[Y] = n\sigma^2, \quad SD[Y] = \sigma\sqrt{n}$$

And $\cancel{Z}$ $\dfrac{Y - E[Y]}{SD[Y]} = \dfrac{Y - n\sigma^2}{\sigma\sqrt{n}} \approx Z$

Chi-Square Distribution

For n standard normal distributions, $Z_1, Z_2, \dots, Z_n$, where $Z_i \sim N(0,1)$, X follows a chi-square distribution if

$X = Z_1^2 + Z_2^2 + \dots + Z_n^2$. Thus $X \sim \chi^2_n$ and it has n degrees of freedom.



$$P[X \geq \chi^2_{\alpha,n}] = \alpha$$

critical value

For a normal population,

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

t - Distribution

t - distribution is used when σ is unknown and we replace it with S.

Derivation

$$Z = \frac{\sqrt{n}\,(\bar{X}-\mu)}{\sigma}$$

$$\cancel{Z = \frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}}}$$

Since σ is unknown, replace with sample sd, S

$$T_n = \frac{\sqrt{n}\,(\bar{X}-\mu)}{S}$$

$$T_n = \frac{\sqrt{n}\,(\bar{X}-\mu)/\sigma}{S/\sigma}$$

$$T_n = \frac{Z}{S/\sigma}$$

we know

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$\Rightarrow \frac{S}{\sigma} = \sqrt{\frac{\chi^2_{n-1}}{(n-1)}}$$

$$\boxed{T_{n-1} = \frac{Z}{\sqrt{\frac{\chi^2_{n-1}}{n-1}}}}$$

$$\frac{\sqrt{n}(X-\mu)}{S} = \frac{Z}{\sqrt{\chi^2_{n-1}/(n-1)}}$$

Chapter 7 : MLE

Parameters :

Ex : $X \sim \text{Poisson}(\lambda)$

Parameter

unknown parameters are represented by θ

Estimator : Formula or statistic used to estimate an unknown ~~parament~~ parameter. Ex : $\hat{\mu} = \bar{X}$

estimator

sample : 2, 4, 6

$\bar{x} = 4$

estimate

MLE :

- Estimation of an unknown parameter to maximise the probability of observed data. Example scenario :

given a set of data : $x_1, x_2, \ldots, x_N$ with unknown mean, μ ;

Estimating μ involves maximizing $L(\mu)$ where

$$L(\mu) = \underset{\substack{\uparrow \\ \text{probability} \\ \text{density function}}}{f(x_1 ; \mu)}\, f(x_2 ; \mu)\, f(x_3 ; \mu) \ldots f(x_N ; \mu)$$

$\hat{\mu}$ = value of μ that maximizes $L(\mu)$

NOTE : for Gaussian distribution $\hat{\mu}$ turns out to be :

$$\boxed{\hat{\mu} = \frac{1}{N} \sum_{n=1}^{N} x_n}$$

MLE of a Bernoulli Parameter

n trials (ind.) each with successful prob. p.

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{o/t.} \end{cases}$$

$$P[X_i = x] = p^{x_i}(1-p)^{1-x_i}$$

$$L(p) = P[X_1 = x_1] \dots P[X_n = x_n]$$

$$L(p) = p^{x_1}(1-p)^{1-x_1} \dots p^{x_n}(1-p)^{1-x_n}$$

$$L(p) = p^{\sum x_i}(1-p)^{n-\sum x_i} \quad \text{where } \sum_i x_i = \text{no. of succ.}$$

$$\ln L(p) = \sum_i x_i \ln(p) + (n - \sum_i x_i)\ln(1-p)$$

$$\Rightarrow \frac{d}{dp} \ln L(p) = \frac{\sum_i x_i}{p} - \frac{n - \sum_i x_i}{1-p}$$

Set the derivative to 0 to find $\hat{p}$

$$\Rightarrow \frac{\sum_i x_i}{\hat{p}} = \frac{n - \sum_i x_i}{1-\hat{p}}$$

$$\Rightarrow \hat{p}(n - \sum_i x_i) = (1-\hat{p})\sum_i x_i$$

$$\Rightarrow \sum_i x_i = n\hat{p} \quad \Rightarrow \boxed{\hat{p} = \frac{\sum_{i=1}^{n} x_i}{n}}$$

MLE of a Poisson Parameter

$$P[X=x] = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$L(\lambda) = P[X_1=x_1] \ldots P[X_n=x_n]$$

$$L(\lambda) = \frac{e^{-\lambda}\lambda^{x_1}}{x_1!} \ldots \frac{e^{-\lambda}\lambda^{x_n}}{x_n!}$$

$$= \frac{e^{-n\lambda}\lambda^{\sum x_i}}{x_1! \ldots x_n!}$$

$$\ln(L(\lambda)) = -n\lambda\ln(e) + \sum x_i \ln(\lambda) - \ln(x_1! \ldots x_n!)$$

$$\ln(L(\lambda)) = -n\lambda + \sum x_i \ln(\lambda) - \ln\left(\prod_{i=1}^{n} x_i!\right)$$

$$\frac{d}{d\lambda}\ln(L(\lambda)) = -n + \frac{\sum x_i}{\lambda} - 0$$

$$\Rightarrow -n + \frac{\sum x_i}{\hat{\lambda}} = 0$$

$$\Rightarrow \frac{\sum x_i}{\hat{\lambda}} = n$$

$$\Rightarrow \boxed{\hat{\lambda} = \frac{\sum x_i}{n}}$$

Ex: 7.2d

For a poisson varr,

$$\hat{\lambda} = \frac{\sum_i x_i}{n} = 2.7$$

$$X \sim \text{Poisson}\ (2.7)$$

$$P[X \leq 2] = \sum_{x=0}^{x=2} \frac{e^{-2.7}(2.7)^x}{x!}$$

$$= 0.4936 \text{ (Ans)}$$

MLE of a Normal Population with 2 unknown parameters

$X_i \sim N(\mu, \sigma^2)$, both μ and σ^2 are unknown

$$f(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_i-\mu)^2}{2\sigma^2}\right]$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left[-\frac{(x_i-\mu)^2}{2\sigma^2}\right]$$

$$L(\mu,\sigma^2) = \left(\frac{1}{2\pi}\right)^{\frac{n}{2}} \times \frac{1}{\sigma^n} \times \exp\left[-\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}\right]$$

$$\ln(L(\mu,\sigma^2)) = \frac{n}{2}\ln\left(\frac{1}{2\pi}\right) - n\ln(\sigma) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}\ln(e)$$

$$\Rightarrow \ln(L(\mu,\sigma^2)) = -\frac{n}{2}\ln(2\pi) - n\ln(\sigma) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

first diff wrt μ,

$$\frac{d}{d\mu}\left(L(\mu,\sigma^2)\right) = \frac{2}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)$$

$$= \frac{1}{\sigma^2}\sum_{x=1}^{n}(x_i-\mu)$$

$$= \frac{1}{\sigma^2}\sum_{x=1}^{n}(x_i-\hat{\mu}) = 0$$

$$\Rightarrow \sum_{x=1}^{n}(x_i-\hat{\mu}) = 0$$

$$\Rightarrow \sum x_i - n\hat{\mu} = 0$$

$$\Rightarrow \boxed{\hat{\mu} = \frac{\sum x_i}{n}}$$

Now diff wrt to σ or σ^2,

$$\frac{\partial}{\partial\sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum(x_i-\mu)^2$$

$$\Rightarrow \frac{1}{\hat{\sigma}^3}\left(\sum(x_i-\hat{\mu})^2\right) = \frac{n}{\hat{\sigma}}$$

$$\Rightarrow \sum(x_i-\hat{\mu})^2 = n\hat{\sigma}^2$$

$$\Rightarrow \hat{\sigma}^2 = \boxed{\frac{\sum(x_i-\hat{\mu})^2}{n}}$$

$$\Rightarrow \boxed{\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2} \quad \text{, since } \hat{\mu} = \bar{x}$$

Ex: 7.2f (pg 17)

Taking the natural logs of the samples:

$\rightarrow$ 0.788, 1.22, 0.470, −0.223, 0.993, 1.194, 0.470, 1.030, 0.916, 0.642

$\hat{\mu} = \bar{x} = 0.7504$

$s_x = 0.4351$

$X \sim N(0.7504, 0.4351)$

$$P[2 < X < 3] = P[0.693 < \log X < 1.099]$$

$$= P[-0.131 < Z < 0.8]$$

$$= \phi(0.8) - \phi(-0.131)$$

$$= \phi(0.8) \cancel{+ \phi(0.13)} - (1 - \phi(1.31))$$

$$= 0.3405$$

MLE of a Uniform Distribution

$$X_i \sim \text{Uniform}(0, \theta)$$

$$f(x_i) = \begin{cases} 1/\theta, & 0<x<\theta \\ 0; & o/t \end{cases}$$

$$L(\theta) = \frac{1}{\theta^n} \quad \text{where} \quad \theta \geq \max(x_1, \dots, x_n)$$

As θ increases $L(\theta)$ decreases. So minimizing

θ maximises $L(\theta)$ so ~~to obe~~

$$\boxed{\begin{aligned} \hat{\theta} &= \max(x_1, \dots, x_n) \\ \hat{\mu} &= \frac{\hat{\theta}}{2} \end{aligned}}$$